

Applying knowledge of rotation and true bearings

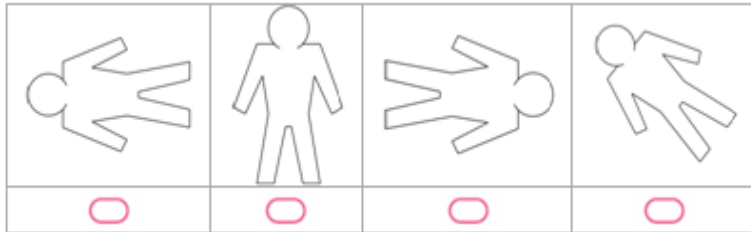
Teacher resource

Question 1

The diagram below is rotated.



Which of the following shows the diagram after a clockwise rotation of 270° ?



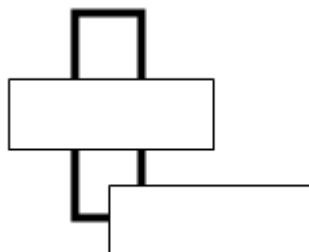
Answer: C

Skill: Students identify a rotation.

Background information

Students should understand that, if a figure has rotational symmetry, each point on the figure, except the point of rotation, can be moved to a different position and the figure will look identical.

Students should also understand that the position of the point of rotation will affect the final position of the rotated figure, although it will not affect its orientation. The diagram below shows the original figure in bold. Although both images resulted from a 90° clockwise rotation, the images are different because different points of rotation were used.

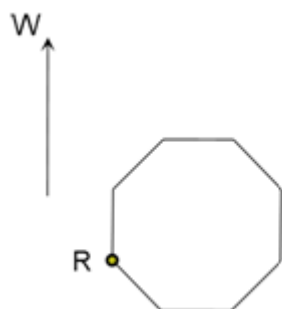


Additional questions

1. Describe diagram C as an anticlockwise rotation of the initial diagram.
2. Describe the other three diagrams as both clockwise and anticlockwise rotations of the initial diagram.

Question 2

The diagram below shows a regular octagonal pathway around a lake.



Robyn walks from point R around the lake. She starts walking in a westerly direction. When she is one third of the way around the lake, in which direction will she be facing?

Answer: North

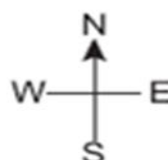
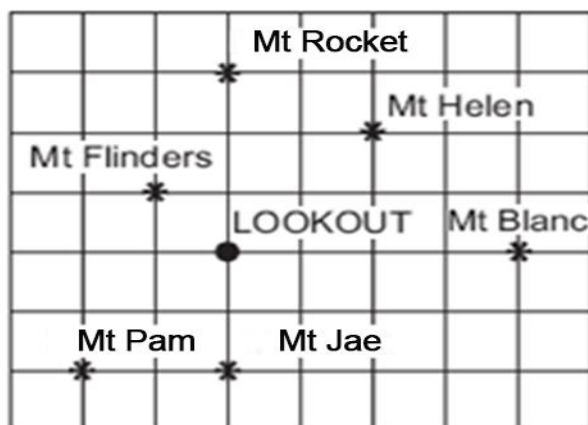
Skill: Students recognise a rotation in a familiar life context.

Additional questions

1. What direction will Robyn be facing when she is two thirds of the way around the lake?
2. How would the answers change to the questions above if Robyn had walked anticlockwise around the lake?

Question 3

The map shows six mountains and a lookout in a national park.



KEY	
*	Mountain (Mt)
●	LOOKOUT

Jim is standing at the lookout, facing south. He turns 225° in an anti-clockwise direction. Which mountain is Jim now facing?

Mt Pam

Mt Flinders

Mt Rocket

Mt Helen

Answer: B

Skill: Students apply knowledge of rotation and bearings.

Additional questions

1. Barney is at the lookout facing Mt Helen. He turns in a clockwise direction to face south. Through how many degrees must he turn?
2. Billie is at the lookout facing south. She turns in a clockwise direction to face Mt Flinders. Through how many degrees must she turn?
3. Anna is at the lookout facing Mt Flinders. She turns in an anticlockwise direction to face Mt Helen. Through how many degrees must she turn?

Question 4

Peter is standing 20 metres due north of a tower. He is facing north.
Carol is standing 20 metres due west of the tower.

Peter turns in a clockwise direction to face Carol.

How many degrees does Peter turn through?

45°
☐

90°
☐

135°
☐

225°
☐

Answer: D

Skill: Students visualise and apply knowledge of rotations.

Additional questions

1. What is the bearing of Peter from Carol?
2. What is the bearing of the tower from Carol?

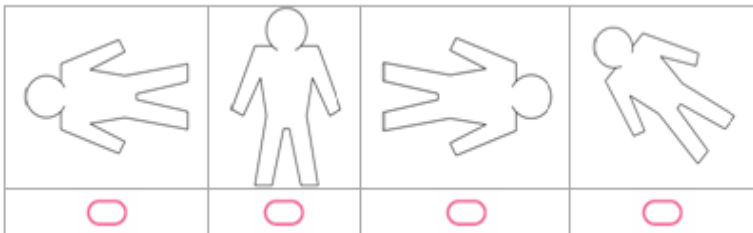
Student activity: Applying knowledge of rotation and true bearings

Question 1

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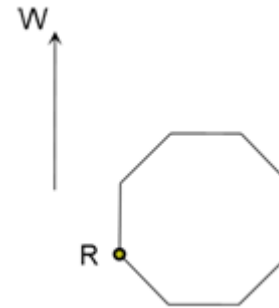


Which of the following shows the diagram after a clockwise rotation of 270° ?



Question 2

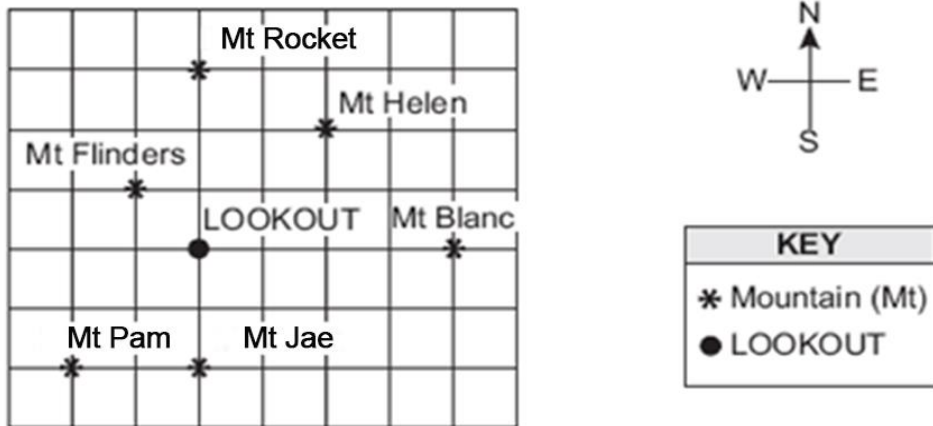
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Which mountain is Jim now facing?

Mt Pam ☐
 Mt Flinders ☐
 Mt Rocket ☐
 Mt Helen ☐

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45° ☐
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 225° ☐